## Wall time per step: Parse 1M PBMC, full (684,932 cells)

Seconds per step. Colour is scaled per step and size: fastest framework 0, slowest 1, on square−root seconds.

| | bixverse mtx | bixverse h5ad | bixverse GPU* | bixverse MLX* | Scanpy mtx | Scanpy h5ad | Scanpy dask | scrapper mtx | scrapper h5ad | scrapper HDF5 | Seurat std mtx | Seurat std h5ad | Seurat BPCells | CyteArc mtx | CyteArc h5ad |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| load | 98 | 48 | 43 | 41 | OOM | OOM | 40 | OOM | OOM | 731 | OOM | OOM | 142 | 384 | 143 |
| qc | 3.1 | 3.2 | 2.1 | 2.0 | – | – | 7.4 | – | – | 13 | – | – | 9.4 | 0.1 | 0.1 |
| normalise | n/a | n/a | n/a | n/a | – | – | 0.0 | – | – | 0.0 | – | – | 4.6 | 27 | 28 |
| hvg | 2.0 | 2.0 | 1.7 | 1.6 | – | – | 8.5 | – | – | 27 | – | – | 27 | 33 | 32 |
| pca | 4.3 | 4.0 | 2.1 | 1.7 | – | – | 17 | – | – | 106 | – | – | 696 | 38 | 41 |
| harmony | 3.7 | 3.8 | 2.3 | 1.1 | – | – | 4.5 | – | – | 255 | – | – | 217 | 228 | 240 |
| knn_snn | 24 | 25 | 13 | 12 | – | – | 33 | – | – | 61 | – | – | 70 | 167 | 172 |
| cluster | 8.7 | 8.8 | 6.9 | 6.6 | – | – | 30 | – | – | 217 | – | – | 256 | 17 | 17 |
| cluster_fast | 37 | 37 | 4.8 | 3.9 | n/a | n/a | n/a | – | – | 330 | n/a | n/a | n/a | n/a | n/a |
| umap | 52 | 51 | 5.8 | 5.2 | – | – | 349 | – | – | 262 | – | – | 454 | 588 | 588 |
| dge | 5.1 | 5.3 | 4.6 | 4.3 | – | – | – | – | – | TIMEOUT | – | – | 1,137 | 39 | 40 |
| tsne_default | 372 | 376 | 323 | 321 | – | – | 1,941 | n/a | n/a | n/a | – | – | 1,606 | n/a | n/a |
| tsne_fft | 116 | 115 | 20 | 31 | – | – | 247 | n/a | n/a | n/a | – | – | 232 | n/a | n/a |
| tsne_quick | 82 | 83 | 41 | 42 | n/a | n/a | n/a | – | – | – | n/a | n/a | n/a | n/a | n/a |
| total (no tSNE) | 202 | 151 | 81 | 75 | failed | failed | 490 | failed | failed | failed | failed | failed | 3,012 | 1,521 | 1,300 |

Black: n/a (step not in that framework), OOM or TIMEOUT where the run died, − not reached.
Each step is scaled over the runs that finished it: at 1M the bixverse CPU tSNE is red because the others timed out.
Scanpy dask has no DGE step. bixverse GPU and MLX run natively without a memory cap.
bixverse CPU tSNE builds its kNN with IVF from 500k (kmknn at 100k). (qd) steps are not in the total.